

Automatic Detection of Epileptic Seizures through CNN–LSTM Analysis of EEG Signals

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Abstract- Despite significant advances in medical science, early diagnosis of epilepsy related seizures remains challenging, with misdiagnosis rates reported to be as high as 30%. Such errors or delays in diagnosis often lead to severe health complications, increased risks, and a reduced quality of life for patients. In this study, a framework is presented for seizure detection using Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM). Spatial features from EEG signals are captured using a Convolutional Neural Network (CNN), while temporal dependencies are modeled through a Long Short-Term Memory (LSTM) network. The Bonn EEG dataset is considered for training and evaluation. The results demonstrated that the proposed model achieved 89.40% accuracy, outperforming Support Vector Machine (SVM), Random Forest (RF). The findings suggest that the proposed approach improves seizure detection and can support better clinical decision-making.

Keywords – Epilepsy detection, Electroencephalography, Long Short-Term Memory

I. INTRODUCTION

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal brain activity. Despite advances in medical science, misdiagnosis rates remain as high as 30%, leading to severe health risks. EEG provides a non-invasive way to capture brain activity, but manual interpretation and conventional machine learning methods often fail to capture both spatial and temporal patterns. This motivates the use of deep learning, particularly CNN–LSTM models, for more accurate seizure detection. Earlier approaches often depended on manual interpretation of EEG signals or conventional machine learning approaches. However, these methods do not capture the spatial with temporal patterns in EEG data. To address these issues, deep learning models are applied.

In the proposed work, CNN and LSTM have shown strong potential in biomedical applications. CNNs are effective in extracting spatial patterns, while LSTMs, are suitable for learning temporal dependencies in signals. By combining these models, EEG-based seizure detection can be made more accurate and reliable. In this study, CNN and LSTM models are applied for automated epilepsy detection using EEG signals. The results presented improved performance, highlighting the role of deep learning to support better diagnosis.

The structure of the paper is: Section II provides an overview of literature review. Section III explains the proposed work methodology. Section IV provides study of results and Section V concludes the work.

II. LITERATURE SURVEY

Acharya *et al.* [1] implemented entropy-based features with SVMs, demonstrating moderate accuracy in distinguishing epileptic conditions. Similarly, Liu *et al.* [2] employed wavelet-based features in conjunction with an SVM classifier, showing improved performance on the Bonn EEG dataset. The emergence of deep learning revolutionized EEG-based seizure detection by enabling automatic feature learning. Zhou *et al.* [3] introduced a CNN model trained directly on raw EEG segments to classify ictal, interictal, and preictal states, reducing the necessity for manual preprocessing. Tuncer and Bolat [4] extended this approach by leveraging bidirectional LSTM (Bi-LSTM) networks on the Bonn dataset, achieving robust classification results through better temporal modeling. Ullah *et al.* [5] demonstrated the efficiency of CNN–LSTM hybrid models, highlighting the advantages of combining spatial and sequential learning in seizure detection framework. Galanopoulou *et al.* [6] proposed a one-dimensional CNN–LSTM architecture tailored for EEG signal analysis, demonstrating substantial improvement in detection accuracy. Geng *et al.* [7] transformed EEG signals using the Stockwell transform before applying a Bi-LSTM, indicating strong performance gains in detection tasks. In a similar hybrid vein, Nie *et al.* [8] combined FFT-based feature extraction with a fully convolutional network nested inside an LSTM, showing promising classification capability.

Beyond CNN–LSTM approaches, alternative deep architectures have been explored. O'Shea and colleagues [9] presented a fully convolutional network for neonatal related seizure detection using raw multi-channel EEG, showing parity with classical SVM-based methods but with full automation. Talathi [10] investigated gated RNNs (GRUs) for early seizure detection, achieving near-perfect accuracy and rapid detection within seconds of seizure onset. Notably, hybrid CNN-SVM models with mutual information-based feature selection was proposed by Hassan *et al.* [11], yielding high classification performance across both Bonn and CHB-MIT datasets. Further work in seizure prediction and detection expanded the repertoire of model architectures and datasets. Golmohammadi *et al.* [12] introduced a recurrent convolutional network for detection using large clinical EEG

databases like TUH, illustrating the necessity of integrating both spatial and temporal contexts.

Asif *et al.* [13] created SeizureNet, employing multi-spectral feature embeddings to classify seizure types, but noted challenges in generalization across EEG datasets. Contemporary deep learning trend reviews have been compiled by Acharya *et al.* [14], illustrating the field's evolution and challenges systematically. Newer methods have also explored advanced signal representations and network designs. Qin *et al.* [15] combined improved variational mode decomposition with deep forest models for EEG classification, while Sarwat and Chahal [16] developed a CNN-LSTM system that achieved exceptionally high accuracy. Chen *et al.* [17] applied feature fusion and CNN for EEG related seizure detection with high accuracy. In addition, recent hybrid models integrating dual-stream networks have been proposed by richhariya and Tanveer [18] using PCA, ICA, and DWT preprocessing methods in combination with CNN-LSTM pipelines, holding promise for future multimodal approaches[22].

III. METHODOLOGY

A. CNN

CNNs are widely implemented for extracting spatial characteristics from biomedical signals. In the context of EEG-based seizure detection, CNNs are effective in identifying local patterns in the signal that may correspond to abnormal brain activity. The core operation of a CNN is the convolution, which applies learnable filters over the input signal to produce feature maps.

The convolution operation is expressed as:

$$y_{i,j}^{(k)} = f\left(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x_{i+m,j+n} \cdot w_{m,n}^{(k)} + b^{(k)}\right) \quad (1)$$

Where $y_{i,j}^{(k)}$ represents the output feature map at position (i, j) for the k^{th} filter, x is the input, $w^{(k)}$ denotes the kernel weights of the k^{th} filter of size $M \times N$, $b^{(k)}$ is the bias term, and $f(\cdot)$ is the non-linear activation function such as ReLU.

Subsequent to convolution, pooling layers are applied to decrease the dimensionality while preserving important features. A common pooling function is max pooling, which is written as:

$$y_{i,j} = \max_{(m,n) \in R} x_{i+m,j+n} \quad (2)$$

where R denotes the local receptive field. This operation ensures translation invariance and decreases computational complexity. Finally, the features are passed through fully connected layers and a softmax classifier for prediction. The softmax function is expressed as:

$$P(y = c | z) = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}} \quad (3)$$

where $P(y = c | z)$ represents the probability of the input belonging to class C , z is the feature vector, and C is the count of classes.

B. Long Short-Term Memory LSTM

LSTMs extend traditional recurrent architectures through the inclusion of gating mechanisms that regulate memory updates. The equations governing the operations of an LSTM cell are:

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (4)$$

Input gate and candidate state:

$$\begin{aligned} i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C[h_{t-1}, x_t] + b_C) \end{aligned} \quad (5)$$

Cell state update:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (6)$$

Output gate and hidden state:

$$\begin{aligned} o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned} \quad (7)$$

Long Short-Term Memory (LSTM) is a type of recurrent neural network that effectively captures long-term dependencies in sequential data. At each time step t , the input x_t is combined with the previous hidden state h_{t-1} to update the network. The forget gate (f_t) controls which parts of the previous cell state C_{t-1} should be retained or discarded, while the input gate (i_t) decides what new information, transformed into candidate values $\tilde{C}_t$, should be added. The updated cell state C_t acts as the memory of the network, carrying essential information across time steps. Finally, the output gate (o_t) determines which parts of the cell state are used to generate the new hidden state h_t . This gated structure allows LSTMs to preserve relevant information over long sequences while mitigating vanishing gradient issues. Through this mechanism, LSTMs retain long-term dependencies making them appropriate for EEG-based seizure detection.

C. Proposed Methodology

The Bonn University EEG dataset was considered in the study. Signals were pre-processed using filters, segmented into windows, and normalized. Features were extracted by CNN and temporal patterns were modeled by LSTM, followed by softmax classification. The model was trained and evaluated. The results were compared with the SVM and RF classifiers.

The following are the phases in the proposed methodology :

Step 1: Dataset Preparation

The Bonn University EEG dataset is considered for the study. It contains normal, interictal, and ictal (seizure) EEG recordings. The dataset is separated into training and testing groups. Ground truth labels (seizure / non-seizure) are provided based on expert annotations.

Step 2: Pre-processing

The EEG signals are filtered with a band-pass filter (0.5–70 Hz) to remove noise and a notch filter to eliminate power line interference (50/60 Hz). Signals are segmented into 2–5 second windows and normalized to ensure consistency across subjects.

Step 3: Feature Extraction with CNN

The CNN automatically extracts spatial patterns and frequency-related features from the signals, reducing the need for manual feature engineering.

Step 4: Temporal Modelling with LSTM

The feature embeddings from the CNN are passed into a LSTM model. The LSTM captures sequential dependencies and temporal variations in EEG signals.

Step 5: Classification Layer

The outputs from the LSTM are passed through fully connected dense layers with a softmax classifier. Each EEG window is classified as either seizure or non-seizure.

Step 6: Model Training

The CNN-LSTM model is trained by Adam optimizer for categorical cross-entropy loss. Dropout is applied to prevent overfitting and enhance generalization.

Step 7: Performance Evaluation

The model performance is evaluated on the test EEG signals from the dataset. Metrics such as accuracy, recall, F1-score, precision and AUC are computed. The CNN-LSTM model is compared with algorithms such as Support Vector SVM and RF.

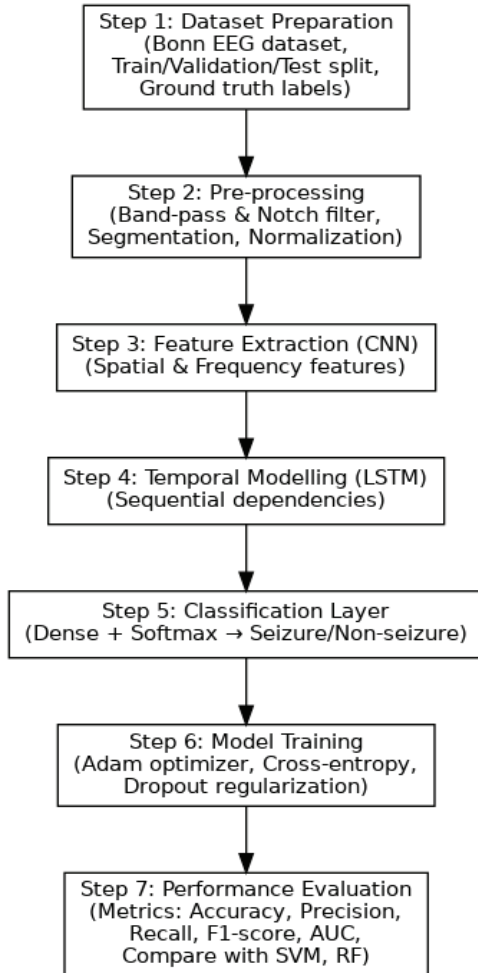


Fig. 1. Flow of Proposed CNN-LSTM seizure detection model

IV. RESULT ANALYSIS

A. Bonn University EEG dataset

The Bonn University EEG dataset[21] has been broadly used as a benchmark for seizure detection studies due to its structured design and reliability. The dataset consists of single-channel EEG recordings sampled at 173.61 Hz with 12-bit resolution, where each segment has a duration of 23.6 seconds, corresponding to 4097 data points. A total of five subsets, labeled A through E, as shown in the fig 2, are included, with each subset containing 100 segments. Sets A and B characterize normal brain activity recorded from healthy volunteers with eyes open and closed, respectively. Sets C and D comprise of seizure-free recordings from epilepsy patients, obtained from both the hippocampal formation of the opposite hemisphere and the epileptogenic zone. Set E includes recordings captured during seizure episodes from the epileptogenic zone. This clear separation between normal, interictal, and ictal conditions makes the dataset suitable for training and evaluation of seizure detection systems[24].

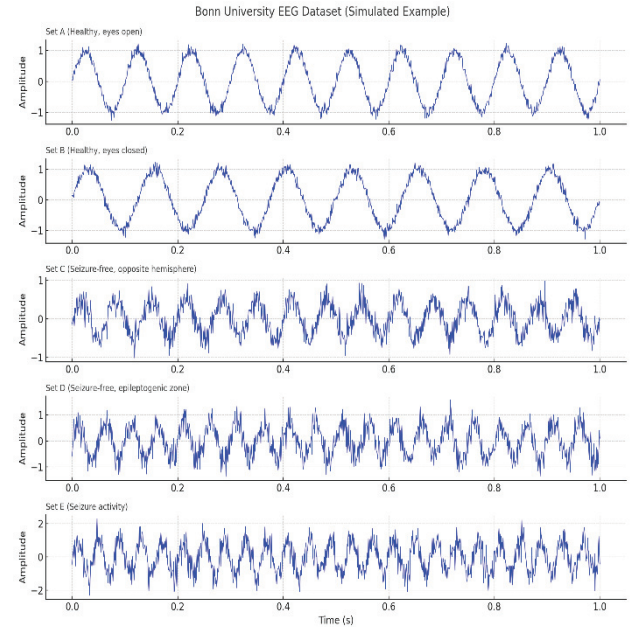


Fig. 2. Bonn EEG Dataset, subsets A - E

B. Experimental Analysis

The comparative performance of the models on the EEG dataset has been summarized in Table 1 and depicted in Fig. 2. It is observed that, SVM and RF achieved accuracies of 78.40% and 81.30%, respectively. Although these models provided a reasonable level of classification performance, their ability to capture complex spatial and temporal dynamics in EEG signals was limited. CNN achieved 85.20% accuracy, while the Long LSTM model obtained 86.10% accuracy. Improved performance was observed with the proposed CNN-LSTM model, which reached an accuracy of 89.40% along with the highest precision, recall, and F1-score among all methods.

The confusion matrix of the proposed model is shown in Fig.4. Out of 90 non-seizure segments, 81 were correctly classified, while 9 were misclassified as seizure. Similarly, 30 seizure segments were correctly identified. These results showed that the model achieved even performance across both the classes, with relatively less false positive and less negative rate. The Area Under the Curve (AUC) of 90.10% further

confirmed the robustness of the proposed approach. The ROC curves of the models were plotted, and their AUC scores were compared in Fig.4. It was observed that the proposed model achieved the highest AUC (90.1%), followed by LSTM (88.4%), CNN (87.1%), Random Forest (83.4%), and SVM (80.5%).

TABLE I. PERFORMANCE COMPARISON OF MODELS ON BONN EEG DATASET

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)
SVM	78.40	77.20	76.10	76.60	80.50
Random Forest	81.30	80.10	79.50	79.80	83.40
CNN	85.20	84.00	83.50	83.70	87.10
LSTM	86.10	85.30	84.20	84.70	88.40
Proposed Model	89.40	88.50	87.90	88.20	90.10

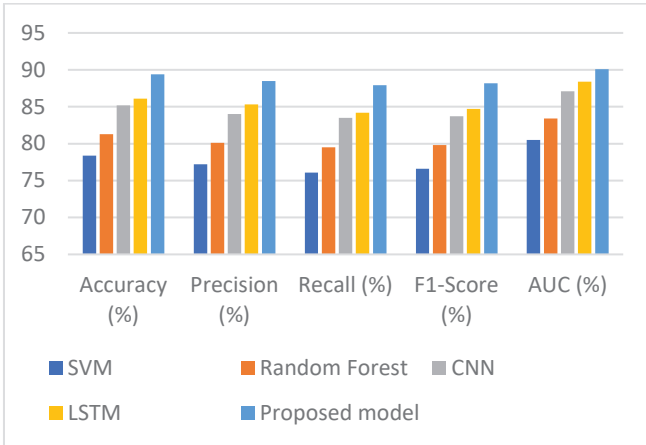


Fig. 3. Performance comparison of different models

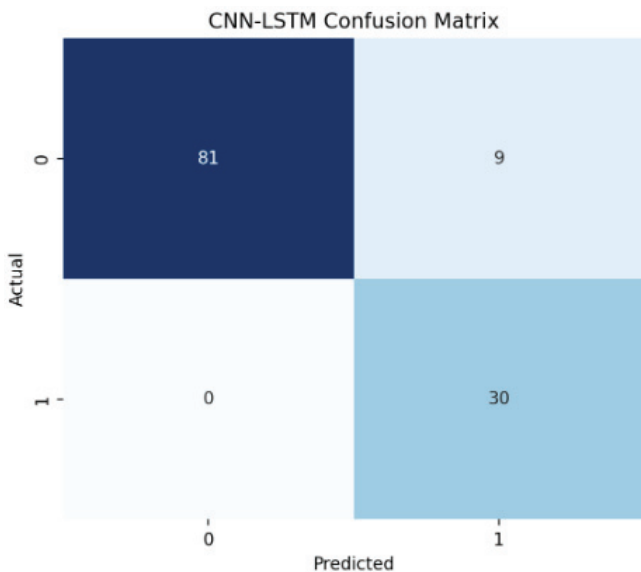


Fig. 4. confusion matrix of proposed CNN-LSTM model

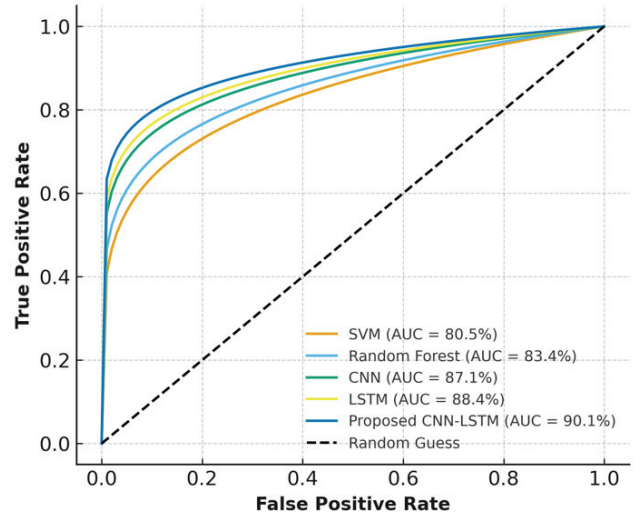


Fig. 5. ROC-AUC comparison of models

V. CONCLUSION

In this study, a framework for epilepsy seizure detection was proposed. SVM and RF achieved moderate results, while deep learning models demonstrated superior performance. CNN effectively extracted spatial features, and LSTM successfully captured temporal dependencies, both contributing to improved classification accuracy. The proposed model achieved the best overall performance, with metric values surpassing those of the individual models. The confusion matrix presented balanced classification between seizure and non-seizure cases. The results determined that the CNN-LSTM framework can serve as a reliable tool for early and accurate epilepsy detection, potentially reducing misdiagnosis rates and improving patient care. With further validation on larger and more diverse datasets, this approach could be integrated into clinical decision-support systems. Future work can explore the inclusion of multimodal data, such as MRI with EEG. Combining different modalities may provide complementary information, thereby enhancing the robustness of seizure detection systems.

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DOI: 10.1109/EICT54103.2021.9733477 (Note: peripheral, but shows CNN usage)

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